

# Fertility and Migration

## The Legacy of the Opioid Epidemic

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Preliminary. Please do not circulate.

### Abstract

In this paper, we examine how the opioid epidemic affected family formation and migration decisions in the United States. Leveraging variation in local exposure to the epidemic—driven by Purdue Pharma’s targeted marketing of OxyContin and proxied by 1996 cancer mortality rates—we find that communities with greater exposure experienced a significant rise in fertility, primarily among unmarried, non-college-educated women in their late twenties. This increase was accompanied by changes in migration patterns: exposure prompted selective out-migration of college-educated women, who tend to have lower expected fertility early in life. Our findings suggest that the epidemic altered local population composition and contributed to long-term demographic divergence across communities.

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# I. Introduction

The opioid epidemic is one of the most devastating and far-reaching public health crises in the history of the United States. It has driven increases in mortality, widespread addiction, economic hardship, and shifts in political preferences (Maclean et al., 2020; Arteaga and Barone, 2025). Over the past three decades, the crisis has evolved from the misuse of prescription opioids to the growing prevalence and use of heroin and fentanyl, leaving a profound impact on the families and communities most directly affected. These effects include greater reliance on welfare and social insurance programs, an increase in the number of children living apart from their parents, higher rates of child maltreatment, and record levels of foster care placements (Evans et al., 2022; Buckles et al., 2022). The crisis has unfolded alongside a broader decline in fertility and a widening divide in family formation patterns by education level. In particular, women without a bachelor’s degree have higher total fertility rates, are more likely to have children at younger ages and outside of marriage, and are significantly less likely to marry than women with a college degree (McLanahan, 2004; Lundberg and Pollak, 2014; Kearney, 2022).

In this paper, we examine how the opioid epidemic altered family formation and internal migration—two demographic processes with important implications for social mobility and economic development. To isolate the effects of the epidemic from confounding socioeconomic trends, we exploit quasi-exogenous variation in local exposure to Purdue Pharma’s 1996 marketing campaign for OxyContin. Our identification strategy leverages detailed features of this campaign, based on unsealed court records from litigation against Purdue Pharma—the manufacturer of OxyContin, the prescription opioid at the center of the crisis. Those records show that at the dawn of the opioid epidemic in 1996, pharmaceutical marketing efforts were concentrated in the cancer pain market with a plan to quickly expand to the much larger non-cancer pain market in those *same* geographic areas. This targeting implied that noncancer patients in high-cancer areas were disproportionately exposed to the opioid epidemic and the unfortunate chain of events that followed. As in Arteaga and Barone (2025), we use cancer mortality in 1996—i.e., before the unfolding of the epidemic—as a measure of exposure. This work and subsequent research, show that mid-nineties cancer mortality is a strong indicator of exogenous exposure to the opioid epidemic.<sup>1</sup>

Our empirical analysis combines data from several sources. First, we use data from the National Vital Statistics System covering the years 1989 through 2018. This dataset provides individual level births and deaths from which we construct fertility outcomes, opioid-related mortality rates and cancer mortality, all measured at the individual level

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<sup>1</sup>This strategy has been used in the literature to study its consequences on health and economic outcomes, see Buckles et al. (2022); Cohle and Ortega (2023); Olvera et al. (2023); Siegal (2023), among others.

with county identifiers. We complement this with individual level data from the Decennial Censuses and the American Community Surveys (ACS), which provide detailed information on demographics, marital status, and migration behavior. Our analysis focuses on 583 commuting zones with populations exceeding 25,000 in 1996, and observed annually from 1989 to 2018, resulting in a panel dataset comprising over 23,000 CZ-year observations.<sup>2</sup>

Our findings reveal that exposure to the opioid epidemic led to an increase in fertility rates. Specifically, we estimate that by 2018, a one-standard deviation increase in exposure raised fertility rates at the commuting zone level by approximately 5.5%. This increase is primarily driven by unmarried women, for whom we detect rising fertility beginning in 2006, with effects that persist and grow over time. Among married women, we estimate fertility increases that are about half as large and statistically significant starting in 2013. The effects are concentrated among women in their late twenties and are higher for those without a college degree.

We next examine whether the observed changes in fertility stem from shifts in individual fertility decisions among individuals who remain in affected areas or are instead driven by changes in the composition of the population within affected communities due to selective migration out of these communities.

We begin with the first channel: changes in individual fertility behavior. Several mechanisms may account for why fertility decisions responded to the opioid epidemic. Among individuals directly affected by the crisis through addiction, reduced contraceptive use may lead to an increase in unintended pregnancies (Terplan et al., 2015). However, we rule out this mechanism as a primary driver of the observed increase in fertility, as we document that opioid use is associated with lower fertility rates.

Another explanation relates to the economic hardship induced by the epidemic, which may have altered the opportunity cost of childbearing. On the one hand, deteriorating socioeconomic conditions may reduce the opportunity cost of time—particularly for women—thus encouraging earlier or increased fertility. On the other hand, a decline in economic resources may limit the affordability of childrearing, thereby reducing fertility. We find that exposure to the epidemic led to declines in female employment, but only among women without a college degree—a group for whom substitution effects are more likely to dominate, leading to an increase in fertility rates (Dehejia and Lleras-Muney, 2004; Perry, 2004; Schaller, 2016).

We then turn to the second channel: changes in the composition of the local population due to migration. The opioid epidemic generated substantial economic hardship—evidenced by increases in SNAP, SSI, and SSDI—which triggered selective out-migration, a key mechanism through which local labor markets adjust to economic shocks (Blanchard et al., 1992; Glaeser and Gottlieb, 2009; Kennan and Walker, 2011). Follow-

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<sup>2</sup>This group of commuting zones accounts for over 97% of the total population in 1996.

ing [Borusyak et al. \(2022\)](#), we use origin-to-destination migration flow data and find that exposure to the epidemic significantly increased out-migration rates, with no detectable effect on in-migration. Beginning in 2005, we document rising out-migration from the most affected communities. These increases continue until 2011, when they stabilize at a 3.5–4% increase in the probability of out-migration per one standard deviation increase in exposure. Those leaving are disproportionately female, single, and more educated (some college or more).

As these outflows accumulated over time, they influenced fertility rates through two channels. First, selective out-migration removed individuals with distinct fertility profiles—specifically, singles and the more educated, who tend to have lower and later completed fertility. Second, migration altered local sex ratios, which in turn affected marriage rates and, consequently, fertility.

Our identification assumption is that, absent the opioid epidemic, areas with higher cancer mortality in 1996 would have followed similar trends in family formation and migration flows as areas with lower cancer mortality. We assess the validity of this assumption using an event study framework, interacting cancer mortality rates with year dummies in a specification that controls for time-invariant CZ characteristics and flexibly adjusts for baseline demographic variables.

We further validate our strategy through two placebo exercises. First, we demonstrate that mid-1990s mortality rates from other causes, do not predict future fertility. Second, we conduct an out-of-sample exercise where we examine whether cancer mortality in 1976 predicts future trends in fertility or other important socio-economic variables, prior to the introduction of OxyContin. We find no evidence of such a relationship. These findings support the interpretation that the observed effects are not due to underlying health trends, but rather to the connection created by the introduction of OxyContin and the unfolding of the opioid epidemic. Finally, we show that our results are not driven by differential exposure to other major economic developments during this period—such as the China shock, the 2001 recession, the great recession, nor are they concentrated in any single region or state.

This paper contributes to three distinct literatures. First, we add to the literature on the socioeconomic determinants of family formation and fertility. A large body of work has documented how economic conditions and individual socioeconomic status shape decisions around fertility. Foundational work by [Becker \(1960, 1973\)](#) emphasized the role of income, opportunity costs, and specialization in determining family behaviors. Subsequent research has shown that economic insecurity—particularly limited labor market prospects for less-educated men—increases nonmarital fertility ([Wilson, 1987, 1996](#); [Black et al., 2005](#); [Autor et al., 2019](#)). This literature has also examined the family formation effects of positive economic shocks to men such as the fracking and coal booms ([Black et al., 2013](#); [Kearney and Wilson, 2018](#)), finding increases in fertility rates with no meaningful

changes in the marriage. Relative to this work, we show how in addition to changes in preferences around fertility, migration is a key under-explored mechanism through which economic shocks shape community level changes in fertility rates and family formation outcomes.

We also extend the literature on the effects of the opioid epidemic by estimating its impact on fertility and migration. Prior research has documented the epidemic’s effects on mortality, addiction, disability claims, municipal finances, housing prices, children’s outcomes, and political behavior (Buckles et al., 2022; Park and Powell, 2021; Cornaggia et al., 2022; Arteaga and Barone, 2025). See Maclean et al. (2020) for a review.

Lastly, we contribute to the literature on the effects of economic shocks on migration by documenting selective migratory responses to the opioid epidemic. These responses vary systematically by demographic and socioeconomic characteristics and stand in contrast to prior work that has found small or insignificant migration responses to negative economic shocks (Autor and Dorn, 2013; Yagan, 2019; Borusyak et al., 2022). Our findings underscore the importance of measuring migration using origin-destination flow pairs disaggregated by group, rather than relying on net migration flows, which may mask meaningful heterogeneity in population responses.

## II. Background: The Unfolding of the Opioid Epidemic

The United States has experienced an unprecedented crisis related to the misuse of and addiction to opioids. As of 2022, over 700,000 lives had been lost to opioid overdoses (CDC, 2023b). During the last decade, a sizeable body of research has studied the origins of the opioid crisis and the factors that shaped its evolution and propagation. This literature has established that the pharmaceutical industry and healthcare providers played a critical role in the origins of the crisis (Eichmeyer and Zhang, 2020; Miloucheva, 2021; Alpert et al., 2022; Arteaga and Barone, 2023). In particular, the aggressive and deceptive marketing of potent opioids with high potential for addiction directed toward physicians, in a setting with financial incentives for doctors to increase prescriptions and with weak monitoring, created the perfect platform for the crisis to unfold.

The beginning of the opioid epidemic is traced to the introduction of OxyContin to the market in 1996 (Quinones, 2015). OxyContin is a prescription opioid manufactured by Purdue Pharma that changed the standard of practice for the treatment of noncancer and nonterminal pain. Prior to the mid-1990s, pain management had focused on cancer and end-of-life pain treatment because of care providers’ fears of the risk of severe addiction (Melzack, 1990). Soon after the introduction of OxyContin, reports of abuse and addiction started to be covered by the media. By 2001, West Virginia, Virginia, Ohio, and Kentucky were pursuing class-action lawsuits against Purdue Pharma and other pharmaceutical companies. During this time, United States Attorneys for Maine and

Virginia, physicians, and community leaders raised concerns to the FDA and Congress about the level of OxyContin abuse (Meier, 2018). The prescription of opioids continued to increase during the 2000s with limited restrictions. At their peak in 2012, opioid prescriptions numbered 81.3 prescriptions per 100 persons (CDC, 2020). The rate of substance use disorder grew by a factor of six between 1999 and 2009 (Paulozzi et al., 2011), and prescription opioid mortality grew by a factor of five (Maclean et al., 2020).

In response to the widespread misuse of prescription opioids and in particular OxyContin, in 2010, Purdue Pharma introduced an abuse deterrent formulation of OxyContin. Evans et al. (2019) and Alpert et al. (2018) show that the reformulation unfortunately led many consumers to substitute toward a dangerous and inexpensive alternative: heroin. As a result, deaths, poisonings, emergency room visits, and enrollments in treatment programs for heroin abuse increased. In particular, between 2010 and 2013, heroin death rates increased by a factor of four with no reduction in the combined heroin and opioid death rate (Evans et al., 2019). From 2013 to this day, the epidemic has been characterized by surging deaths related to the use of synthetic opioids, particularly fentanyl. Fentanyl, an extremely potent synthetic opioid, is more profitable to manufacture and distribute than heroin and has a higher risk of overdose. Indeed, fentanyl-related deaths account for almost the entire increase in drug overdose mortality between 2014 and 2021.

Mortality from opioids is only one of the many social costs associated with the opioid epidemic. Some 10.1 million people aged 12 or older are estimated to have misused opioids in the past year in the US (SAMHSA, 2020). These numbers are orders of magnitude larger than the number of deaths and help rationalize why the opioid epidemic has disrupted health and economic opportunities, affecting individuals and their communities. Notably, the epidemic has increased disability rates and Supplemental Nutrition Assistance Program (SNAP) utilization (Powell et al., 2020; Savych et al., 2019; Arteaga and Barone, 2023) and affected labor force participation and employment rates (Krueger, 2017; Ouimet et al., 2020).

This evidence highlights the profound toll of the opioid epidemic on individuals, families, and communities. While its health and economic consequences are well documented, much less is known about how the epidemic has shaped key demographic behaviors. In the rest of the paper, we examine how exposure to the opioid crisis affected fertility and internal migration decisions—two critical processes that influence long-run community structure, social mobility, and economic development.

### III. Data

#### III.a Mortality

We use restricted-use data from the Detailed Multiple Cause of Death (DMCD) files from 1989 to 2018 to construct mortality measures. These files record every death in the United States, including the county of residence, the underlying cause of death, and up to 20 additional contributing causes, and thus represent a census of deaths in the U.S. The 1989–1998 data use ICD-9 codes to classify causes of death, while the 1999–2018 data use ICD-10 codes. We compute the 1996 cancer mortality rate to measure exposure to the opioid crisis.<sup>3</sup> Figure 1 shows the distribution of cancer mortality across geographies in 1996 and Table 1 shows summary statistics for cancer-mortality in the pre- and post-periods.

We construct a measure of drug-related overdose deaths to study the mortality effects of the epidemic. We define drug-induced mortality to include deaths from poisoning and medical conditions caused by the use of legal or illegal drugs, including prescription opioids, heroin, and synthetic opioids such as fentanyl.<sup>4</sup> We analyze drug-related deaths separately for women and men, and also leverage educational attainment data included in death certificates to document effects by education level—specifically, for individuals with some college or more, and those without any college education.<sup>5</sup> Table 1 presents summary statistics for drug-induced mortality overall and by sex.

#### III.b Fertility Rates

We use the National Center for Health Statistics (NCHS) Natality Files to construct fertility rates. These data represent the universe of births in the United States and include geographic identifiers—such as the county of birth—as well as detailed demographic characteristics of the birthing individual, including age, marital status, and educational attainment. From these data, we compute average fertility rates at the commuting-zone level. The overall fertility rate is defined as the number of births to individuals in a given age group divided by the population in that age group.

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<sup>3</sup>We define cancer deaths as those caused by malignant neoplasms (codes 140–208 in ICD-9 and C00–C97 in ICD-10) and in situ neoplasms, benign neoplasms, and neoplasms of uncertain or unknown behavior (codes 210–239 in ICD-9 and D00–D48 in ICD-10).

<sup>4</sup>Drug-related overdose deaths can be difficult to categorize; for this reason, we favor a broad measure of mortality. This category includes codes E850–E858, E950.0–E950.5, E962.0, and E980.0–E980.5 in ICD-9; and X40–X44, X60–X64, X85, Y10–Y14, and T36–T39, T40–T50 in ICD-10.

<sup>5</sup>The 2018 data were the first for which educational attainment was consistently reported using the 2003 revision of the death certificate across all deaths (unless missing entirely). In prior years, some deaths still used the 1989 revision’s categorization. This change prevents us from constructing more granular education categories. The next section provides more detail on these definitions.



We further disaggregate fertility rates by marital status and educational attainment.<sup>6</sup> For the analysis by marital status, we define the married (unmarried) fertility rate as the number of births to married (unmarried) individuals in a given age group divided by the population of married (unmarried) women in that age group. The numerator is obtained directly from the NCHS Natality Files; the denominator is constructed using data from the U.S. Census and the ACS. Figure 2 displays three decades of fertility trends by age and marital status. Over this period, mothers aged 25–29 consistently accounted for the largest share of births, followed by those aged 20–24 in earlier years and those aged 30–34 after 2010. The number of births to unmarried mothers increased across all age groups, while births to married mothers declined among women under age 30.

To compute fertility rates by educational attainment, we classify mothers into two groups: those with some college or more, and those without any college education. The NCHS data include information on years of schooling, but we reclassify education levels to account for changes in how this information is reported across time and states. We define “some college or more” as having completed more than 13 years of education, any number of years of college without a degree, or any higher degree (e.g., a bachelor’s or master’s degree). If education is reported but does not meet this definition, we classify the individual as having “no college education.” As with the marital status analysis, the denominator is constructed using Census and ACS data.

Table 2 presents cross-sectional regressions examining the demographic and economic determinants of fertility rates in 2000. Fertility rates are disaggregated by marital status and educational attainment. Among demographic characteristics, a higher share of the population aged 50 to 64 is positively associated with unmarried fertility, but negatively associated with married and no-college fertility. In contrast, the share of the population over age 66 shows no meaningful relationship with fertility rates. The white population share is positively associated with married fertility, but negatively with unmarried fertility, whereas the Hispanic population share is consistently positively associated with fertility across all subgroups. A higher female population share is negatively associated with unmarried fertility, but positively associated with married fertility. Opioid-related mortality has a small negative relationship with fertility rates among mothers without any college education, while other adult mortality is positively associated with unmarried fertility, but negatively with married fertility. Cancer mortality, by contrast, shows no statistically significant association with fertility outcomes. Turning to economic variables, the employment rate of men aged 15–44 is strongly and positively associated with fertility, while the employment rate of women has a negative but statistically insignificant relationship. Local poverty rates are positively associated with married fertility, but neg-

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<sup>6</sup>The Natality Detail Files include information obtained from each U.S. birth certificate. There is some variation across states and over time in the availability of specific variables. Table A.xx documents the variables used and reports missingness rates by state. In some cases, we exclude states from specific analyses due to data limitations; we note these exclusions when relevant.



actively associated with fertility among women with some college or more. Overall, with the exception of Hispanic population share and male employment, all other determinants display opposite signs across the married vs. unmarried and college vs. no-college fertility breakdowns. Finally, column 5 of Table 2 presents the cross-sectional determinants of marriage rates among women aged 15 to 44. A higher share of White and Hispanic individuals is associated with increased marriage rates, while a higher share of females is associated with lower marriage rates. Employment rates for men and women have opposite effects: male employment is positively associated with marriage rates, whereas female employment is negatively associated.

### III.c Migration

We collect data on county-to-county migration flows from the IRS Statistics of Income (SOI) Tax Stats. These data are based on the universe of Forms 1040 filed and processed by the IRS during a calendar year and are available for filing years 1990 through 2021. The dataset includes the number of personal exemptions claimed, which approximates the number of individuals in a given county. Migration flows are identified based on year-to-year address changes reported on individual income tax returns. These data capture migration patterns across all U.S. counties and include both inflows—the number of new residents moving into a county and their origin—and outflows—the number of residents leaving a county and their destination. While these data cover most internal migration, they do not include demographic information. To examine migration flows for specific subpopulations, we complement the IRS data with information from the Census and the American Community Survey (ACS). Specifically, we construct the number of moves from each geographic area based on individuals’ reported locations in the previous year and rely on person weights to generate representative estimates.<sup>7</sup> Informed by Foschi et al., 2023, we drop the four least populous states of the contiguous United States (South and North Dakota, Vermont and Wyoming) from our analysis sample due to the volatile state population swings observed in the ACS data.

### III.d Additional Data and Geographic Harmonization

*Demographics.* We use three data sources for population counts. First, the Survey of Epidemiology and End Results (SEER), which reports population at the county level and by age, race, sex, and Hispanic origin. We use these data to construct the denominators for mortality rates, e.g., drug-induced mortality by sex. SEER does not provide population counts by educational attainment. For education-specific outcomes, we leverage data from the Census and the ACS. These data are at the county level and include the

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<sup>7</sup>Unfortunately, the 1990 and 2000 Censuses ask about residence five years prior. Therefore, for Census years, we construct the number of moves based on this retrospective question.

years 1990 and 2000 from Census and 2006 onward from the ACS. When using Census and ACS data to construct denominators, we linearly interpolate population counts.

*Geographic harmonization.* Mortality and natality data are available at the county level; we use the crosswalks developed by [Autor and Dorn \(2013\)](#) to aggregate the data to CZ level. Some CZs cross state borders; when this happens, the CZ is assigned to the state with the larger share of the CZ population. This criterion helps preserve the strong within-cluster and weak between-cluster commuting ties.

In sum, we build a data set at the commuting-zone level, covering the period from 1989 to 2018 for our outcome variables and the measure of exposure to the epidemic. We choose commuting zones as our unit of observation since it is the geographic space that captures one’s economic life—which often spans beyond county borders. We restrict our sample to areas with more than 25,000 residents. This represents 97% of the total population. Our final dataset is a balanced panel of 583 commuting zones and consists of over 23,000 observations. Table 1 presents summary statistics for key variables for our analysis sample, in pre- and post-periods.

## IV. Empirical Strategy

To identify the effect of the opioid epidemic on family formation outcomes, we exploit rich geographic variation in the early promotional efforts of prescription opioids as an exogenous source of exposure. Following the approach in [Arteaga and Barone \(2023\)](#), we use 1996 cancer mortality as a proxy for local exposure to the opioid epidemic. The rationale behind this strategy is that the size of the cancer pain market in the mid-1990s was a key determinant of where pharmaceutical companies initially concentrated their marketing efforts for opioids. A large body of research—and multiple legal rulings—has established a strong link between these early promotional campaigns and the subsequent unfolding of the opioid crisis. We run the following specification over our sample of commuting zones for the period 1989 to 2018:

$$\Delta y_{ct} = \sum_{\tau=1989}^{2018} \phi_{\tau} CancerMR_{c,1996} \mathbf{1}(Year = \tau) + \sum_{\tau=1989}^{2018} \alpha_{\tau} X_{ct_0} \mathbf{1}(Year = \tau) + \gamma_{st} + v_{ct}, \quad (1)$$

where  $c$  indexes the commuting zone,  $s$  the state, and  $t$  the year.  $t_0$  corresponds to 1996—the year of OxyContin’s launch—or the closest year with available data (e.g., 2000 when Census data are used). We define  $\Delta$  as the long-change operator: for any random variable  $W_{ct}$ ,  $\Delta W_{ct} = W_{ct} - W_{ct_0}$ .

The outcome of interest is given by  $y_{ct}$ .  $X_{ct_0}$  is a vector of baseline demographic

variables such as the share female, white and age groups.  $CancerMR_{c,1996}$  is the cancer mortality rate in commuting zone  $c$  in 1996 and it is interacted with a full set of year fixed effects indexed by  $\tau$ . In this specification, the coefficients for the pre-epidemic period; i.e.,  $\phi_{1989}$ ,  $\phi_{1990}$ , to  $\phi_{1995}$ , test whether outcomes in higher and lower cancer mortality areas followed similar trends before 1996. In the result section, we present scaled estimates of the  $\phi_\tau$  coefficients, we refer to these as *effect sizes*. Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in the baseline year  $t_0$ . This yields the percent change in the outcome associated with a one standard deviation increase in exposure.

This research design allows us to control for state-specific trends and state-level policy changes which were common during this period, as well as time invariant commuting zone characteristics. When estimated on a perfectly balanced panel, this specification is equivalent to a regression in levels with commuting zone fixed effects. However, our panel is not perfectly balanced for some outcomes—e.g., fertility by marital status cannot be constructed for California after 2016. In such cases, the specification in Equation (1) is more efficient, and we adopt it as our baseline approach.

## V. Results

### V.a Fertility

We examine how exposure to the opioid epidemic affected fertility patterns across U.S. commuting zones. We start by presenting the evolution of the fertility rate in areas more exposed to the epidemic—those in the top quartile of cancer mortality in 1996—and areas with less exposure. Panel (a) of Figure 3 shows that while fertility rates have generally declined over the past three decades, the decline was less pronounced in areas more exposed to the epidemic. Estimating Equation (1), we find that exposure to the opioid epidemic led to a substantial (relative) increase in fertility among women of reproductive age. Panel (b) of Figure 3 shows that a one standard deviation increase in exposure—proxied by local 1996 cancer mortality rates—led to an approximately 5.5% increase in the overall fertility rate among women aged 15 to 44 by 2018, relative to its 1996 baseline. These effects start to be significant in 2005, and have steadily grown since then. These increases in fertility are driven by women aged 25 to 29. Figure 4 presents a breakdown of the fertility results by age and shows that this is the only group in which we observe a significant effect of exposure to the epidemic.

Panel (a) of Figure 5 presents the effects of exposure on fertility by marital status. We find that a one-standard deviation increase in exposure raised fertility among unmarried women by 4.7% by 2018, compared to a 2.8% increase among married women. The dynamics of these effects also differ meaningfully: the increase in fertility among unmarried

women becomes statistically significant starting in 2005, whereas it takes an additional eight years—until 2013—for the effect on married women’s fertility to reach statistical significance.

When we disaggregate the effects by education level, we find that the increase in fertility is more pronounced among women without any college education, particularly during the decade from 2005 to 2015. Panel (b) of Figure 5 shows that in areas more exposed to the opioid epidemic, fertility among women without a college degree increased by 5% by 2010. In contrast, the increase among women with some college education was only 1.8% over the same period. Over time, fertility rates for both groups continued to rise. By 2018, fertility had increased by 6% among women without any college education and by 5% among those with some college or more.

## VI. Mechanisms

The effects on fertility and family formation that we document may stem either from changes in fertility decisions or from selective migration and mortality that altered the demographic composition of the communities most affected by the epidemic. In the following sections, we explore whether the epidemic shaped the demographic composition of communities. We document effects on migration, mortality and the gender composition of communities.

### VI.a Opioid Use and Fertility

We begin by examining a direct mechanism through which the opioid epidemic could influence individual fertility decisions. Terplan et al. (2015) review the literature on contraceptive use among women with opioid use disorders and find that these women are less likely to use contraception compared to non-drug-using populations. This lower rate of contraceptive use could potentially lead to higher fertility. However, using data from the 2018 National Survey on Drug Use and Health (NSDUH), we find no significant relationship between current opioid use and pregnancy status (Table A1).

Moreover, a back-of-the-envelope calculation suggests that women with opioid use disorders may, in fact, have lower fertility rates. Specifically, the prevalence of opioid use disorder among women aged 15 to 44 is approximately 0.9% of the population (NSDUH, 2020), while births involving neonatal abstinence syndrome account for just 0.6% of all births (CDC, 2023a). Taking into account estimates that approximately 80% of neonates exposed to opioids in utero develop withdrawal symptoms (Hudak et al., 2012), this implies that births to women with opioid use disorders are underrepresented in the birth population. This pattern is consistent with the possibility that women with opioid use disorders experience lower fertility overall.

## VI.b Economic Effects

Next, we turn to mechanisms through which the opioid epidemic may indirectly influence individual fertility decisions via its broader community-level impacts. (Arteaga and Barone, 2023) show that the opioid epidemic induced significant economic hardship, as reflected in increases in the share of the population receiving SNAP, SSI, and SSDI benefits. This economic distress may affect fertility decisions through both income and substitution effects.

On one hand, economic hardship may reduce the opportunity cost of time, thereby encouraging earlier or increased fertility—a substitution effect. On the other hand, declining economic resources may limit the affordability of childrearing, thereby reducing fertility—an income effect. Prior research has shown that for women and lower-income individuals, the substitution effect tends to dominate, whereas for men, the income effect is more pronounced (Perry, 2004; Dehejia and Lleras-Muney, 2004; Schaller, 2016). While data availability constraints limit our ability to assess differential effects of welfare and social insurance take-up by sex, we use aggregate county-level data by sex and education from the Longitudinal Employer-Household Dynamics (LEHD) to compute sex-specific employment impacts of the opioid epidemic.

In Figure 6, we find no detectable effects of the opioid epidemic on employment rates for men. For women, we estimate negative effects, though they are not statistically significant. However, when we split the data by education, we observe that among women without a college education, the opioid epidemic induced a continuous decline in employment. By 2018, a one standard deviation increase in exposure to the epidemic is associated with an 8% reduction in employment for this group. This deterioration in economic conditions lowers the opportunity cost of childbearing, which may help explain the observed increase in fertility among these women.

## VI.c Migration Effects

We begin by estimating the impact of the opioid epidemic on origin-by-destination migration decisions. We estimate the following regression on each annual cross-section of commuting zone pairs for all years  $t$  in our sample:

$$\Delta y_{ol,t} = \phi_{o,t} Cancer_{o96} + \phi_{l,t} Cancer_{l96} + \alpha_{o,t} X_{o96} + \alpha_{l,t} X_{l96} + \gamma_{so} + \gamma_{sl} + v_{ol,t} . \quad (2)$$

Where  $y_{ol,t}$  is the log of the probability of migrating from commuting zone  $o$  to  $l$  in year  $t$ , and  $\Delta y_{ol,t}$  is the change in this variable between year  $t$  and the baseline year ( $t_0$ ).  $Cancer_{o96}$  and  $Cancer_{l96}$  capture exposure to the epidemic in the origin and destination locations, respectively.  $\gamma_{so}$  and  $\gamma_{sl}$  represent origin and destination state fixed effects. We

include demographic controls in both origin and destination commuting zones at baseline ( $X_{o96}$  and  $X_{t96}$ ). This specification follows [Borusyak et al. \(2022\)](#) who suggest including exposure to shocks in origin and destination locations. Intuitively, this captures that moving decisions depend upon both the shock in the current location and the shock in the potential alternative locations.

The main coefficient of interest,  $\phi_{o,t}$ , captures the migration semi-elasticity with respect to exposure to the opioid epidemic. Specifically, holding all else constant, a one-unit increase in cancer mortality in the origin location leads to a  $\phi_{o,t}$  percent change in the probability of out-migrating from  $o$  by year  $t$ , relative to the baseline year. To ease the interpretation, we present scaled coefficients, i.e., we multiply the semi-elasticity by the standard deviation of cancer mortality in 1996.<sup>8</sup>

Figure 7 presents our results. Exposure to the opioid epidemic significantly increases out-migration rates. Panels (a) and (b) present complementary evidence on this pattern. Panel (a) presents estimates of inter-county out-migration effects using IRS data and shows that a one-standard deviation increase in exposure to the epidemic led to a 4% increase in out-migration rates.<sup>9</sup> This estimate reflects moves among individuals of all ages. Panel (b) presents estimates of inter-CZ migration focusing on individuals aged 15 to 44, who are of childbearing age and therefore are our primary sample of interest. In this sample, out-migration effects are more pronounced: a one-standard deviation increase in exposure led to a 9% increase in out-migration rates.

We document significant heterogeneity in migration responses by sex and educational attainment. Panel (c) shows that inter-CZ out-migration responses are driven by women, and Panel (d) reveals that this effect is concentrated among women with some college education. This pattern of differential migration has important implications for fertility, given that educational attainment is a strong predictor of fertility behavior. For instance, using data from the ACS, we find that women without any college education have fertility rates by age 30, that are 48% higher than those of women with some college education.

## VI.d Compositional Changes

With respect to direct health impacts—specifically drug poisoning mortality—men experience significantly higher death rates than women. In 2022, the drug-related mortality

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<sup>8</sup>Our use of cross-sectional regressions in our migration analysis is motivated by computational efficiency. It can be readily shown that estimating Equation 2 over all cross-sections is numerically equivalent to a stacked “event-study style” panel regression, analogous to Equation 1. We therefore continue to present our results in event-study style graphs and maintain the convention of indicating the baseline year with a zero coefficient.

<sup>9</sup>The IRS migration data suppresses county-to-county migration flows below certain thresholds for confidentiality purposes, as a result, differential missingness across CZs makes aggregating to CZ-to-CZ migration flows challenging. At this paper’s current stage, when using the IRS migration data we focus on inter-county migration among county-pairs with 1996 migration rates in the top two quintiles which are unaffected by this suppression.

rate for men was 2.5 times higher than that for women. However, when examining the effects of exposure to the opioid epidemic, we find that the effect sizes are broadly similar across sexes. In Figure 8, we document that by 2016, a one-standard deviation increase in 1996 cancer mortality increased drug mortality among both men and women of approximately 65%. In absolute terms, however, exposure to the epidemic translates into greater increases in the number of drug-related deaths among men than among women.

Next we evaluate whether the documented effects on migration were large enough to alter the demographic composition of the communities most affected by the epidemic. We start by examining the share of females in the population aged 20 to 44 years. In Figure 9, we find that a decade into the crisis, a one-standard deviation increase in 1996 cancer mortality, reduces the share of women aged 20 to 44 by approximately 0.27 percentage points. The literature has documented how unbalanced sex ratios affect economic and marriage decisions, but prediction on fertility are less clear (Wilson, 1987; Chiappori et al., 2002; Angrist, 2002).

## VII. Robustness Checks

In this section, we provide evidence in support of our identification strategy, explore alternative explanations for our findings and test the sensitivity of our results.

### VII.a Evidence on the Exogeneity Assumption and Exclusion Restriction

The validity of the identification strategy requires that, in the absence of prescription opioid marketing, areas with higher cancer mortality in 1996 would have followed the same trends in health and economic outcomes as areas with lower cancer mortality—all along the support of the 1996 cancer mortality distribution. To support this assumption, we present several exercises.<sup>10</sup>

First, we perform an out-of-sample dynamic reduced-form analysis to test if lagged cancer mortality predicts future fertility prior to the onset of the opioid epidemic. That is, we run Equation (1) over a sample of CZs for the years 1989 to 1995 and estimate if cancer mortality in 1976 predicts this outcome. We present the results of this analysis in Panel (b) of Figure 10. These results demonstrate that before the introduction of OxyContin, there is no relationship between our outcome measure and lagged cancer mortality—the estimated coefficients are statistically indistinguishable from zero, and there is no evidence of a pattern.

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<sup>10</sup>In this draft we focus on fertility as the main outcome of analysis. Arteaga and Barone (2023) provide complementary evidence on the validity of this strategy and present results for these tests on mortality, economic, and political outcomes.



We further probe the validity of our design by estimating event-study regressions with placebo instruments—i.e., mid-1990s mortality from causes unrelated to cancer.<sup>11</sup> Finding a good placebo instrument is challenging given that the causes that underlie the incidence of cancer and of other conditions such as heart disease are not independent (Chiang, 1991; Honoré and Lleras-Muney, 2006). As a result, there is substantial overlap across underlying causes, and the correlation across measures is very high, especially among elderly age groups. With this caveat, in Panel (b) of Figure 10, we show placebo instrument regressions for mortality from hypertension and pneumonia, which are less likely to be affected by the previous concern but still capture community-level health trends. We find no relationship between these placebo mortality rates and total fertility.

We also expand the set of covariates to account for health behaviors, access to health-care and economic factors and the correlation of these with cancer. For example, Dong et al. (2022) document that the share of smokers and the share of adults with overweight are among the main risk factors associated with cancer mortality. First, we show that including contemporaneous cancer mortality in our set of controls also does not alter our results (Panel (c) of Figure 10). We then expand the set of controls in our baseline to include the share of smokers, the share of adults with overweight, the share of primary care physicians, and the infant mortality rate, all measured at the CZ level at baseline or the earliest available data point. Panel (c) of Figure 10 presents estimates from a specification where we add these controls, interacting each of these with year dummies. Our results remain unchanged.

Similarly, we include a set of economic covariates to our main specification. We interact each economic control measured at baseline with year dummies; these controls include the unemployment rate, the share of employment in the manufacturing sector, income per capita, and the share of the population that has completed some college.<sup>12</sup> Panel (c) of Figure 10 also shows that the inclusion of these covariates does not affect our main estimations for fertility or migration.

## VII.b Exposure to Economic Shocks

We account for geographic exposure to major economic shocks that could confound the relationship between opioid exposure and our outcomes. For each shock, we include a measure of local exposure interacted with year dummies to flexibly control for its varying geographic impact.<sup>13</sup>

Contemporaneous with the unfolding of the opioid epidemic, the United States economy was affected by rising import competition from China and two major recessions

<sup>11</sup>These placebo instruments are also known as negative instruments (Danieli et al., 2023).

<sup>12</sup>All of these controls are measured at the CZ level at baseline or the first available year.

<sup>13</sup>We follow the literature to construct these measures; (Arteaga and Barone, 2023) provides the details. Correlations with opioid exposure range from  $-0.04$  to  $0.18$ : NAFTA:  $-0.04$ , China shock:  $0.18$ , 2001 recession:  $0.05$ , Great Recession:  $0.02$ .

(2001 and 2007–09. These developments could influence the same family formation and economic outcomes we study.

To address this, we control for exposure the regional import competition measure from [Pierce and Schott \(2020\)](#). Exposure to the 2001 and Great Recession is measured using changes in CZ-level unemployment around each downturn. As shown in Panel (d) of Figure 10, controlling for these economic shocks does not alter our results. The estimates remain stable and quantitatively similar to those from the baseline specification, suggesting that our findings are not driven by these coinciding developments.

### VII.c Alternative Sample Restrictions

Finally, we assess whether our results are sensitive to the exclusion of any single state or driven by specific regions of the country. In Figure 11, we conduct a series of robustness checks in which we sequentially exclude each state from the sample, as well as broader geographic regions—the commuting zones in the Appalachia and the Rust Belt. We find that our results remain robust to these changes in the analysis sample, suggesting that they are not driven by any particular state or region.

## VIII. Discussion

This paper documents how the opioid epidemic, one of the most consequential public health crises in recent U.S. history, shaped two fundamental demographic processes: fertility and internal migration. Using exogenous variation in exposure to the epidemic stemming from Purdue Pharma’s early marketing strategy for OxyContin, we show that more heavily exposed communities experienced a substantial rise in fertility—particularly among younger, unmarried, and less-educated women. At the same time, we find strong evidence of selective out-migration, concentrated among college-educated women and those with lower expected early-life fertility. These compositional shifts—rather than changes in fertility preferences alone—appear to be a key mechanism behind the rise in fertility in exposed areas.

Our findings carry several broader implications. First, they reveal a novel channel through which public health crises can influence long-term demographic and social patterns—not only through direct health effects but also via changes in family formation and population structure. Second, they underscore the importance of considering both behavioral and compositional responses when evaluating the demographic consequences of large-scale shocks. Finally, by showing how the opioid epidemic has contributed to demographic divergence across communities, our results speak to the growing geographic inequality in social and economic conditions within the United States.

Future research could build on our findings by examining how these shifts in family formation and migration interact with other long-run outcomes, such as child develop-

ment, educational attainment, or intergenerational mobility. Additionally, understanding how institutional responses—such as access to addiction treatment, economic support programs, or housing policy—moderate these effects remains an important avenue for further exploration.

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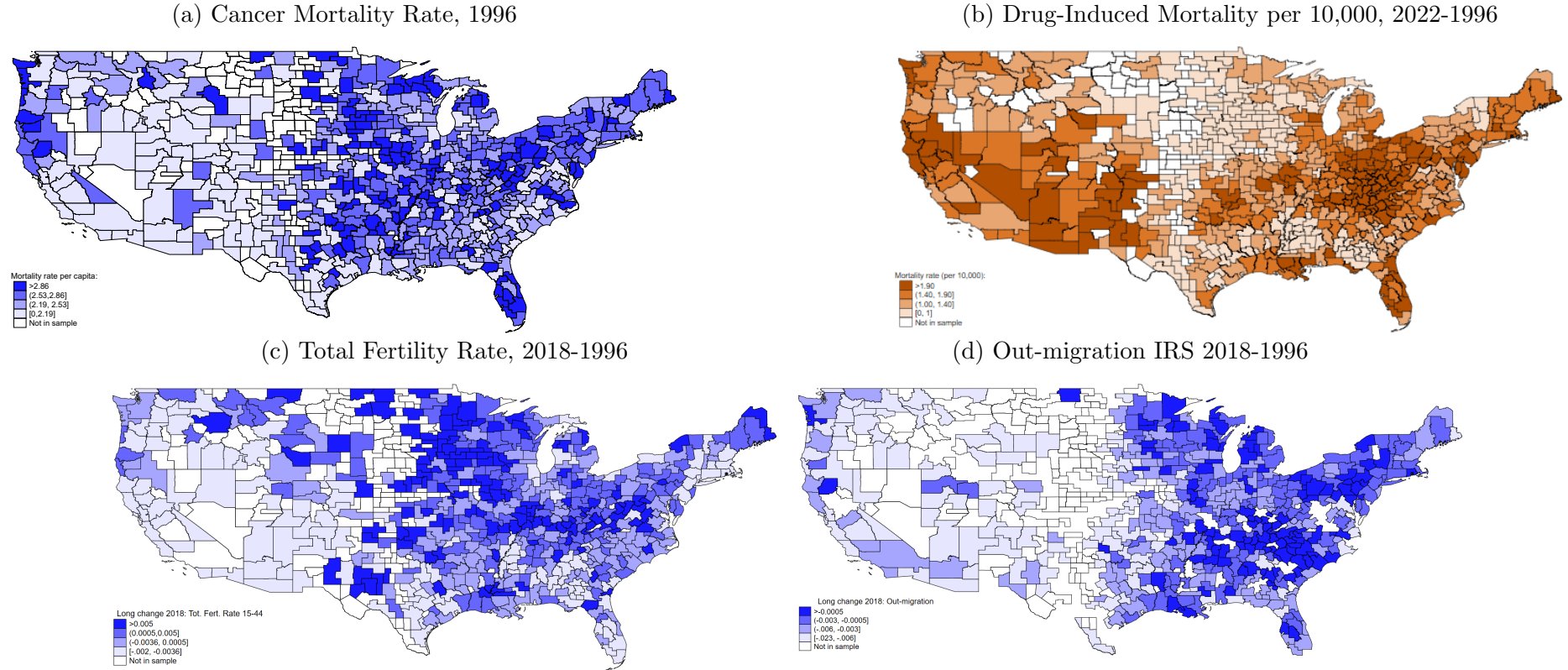
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## IX. Figures

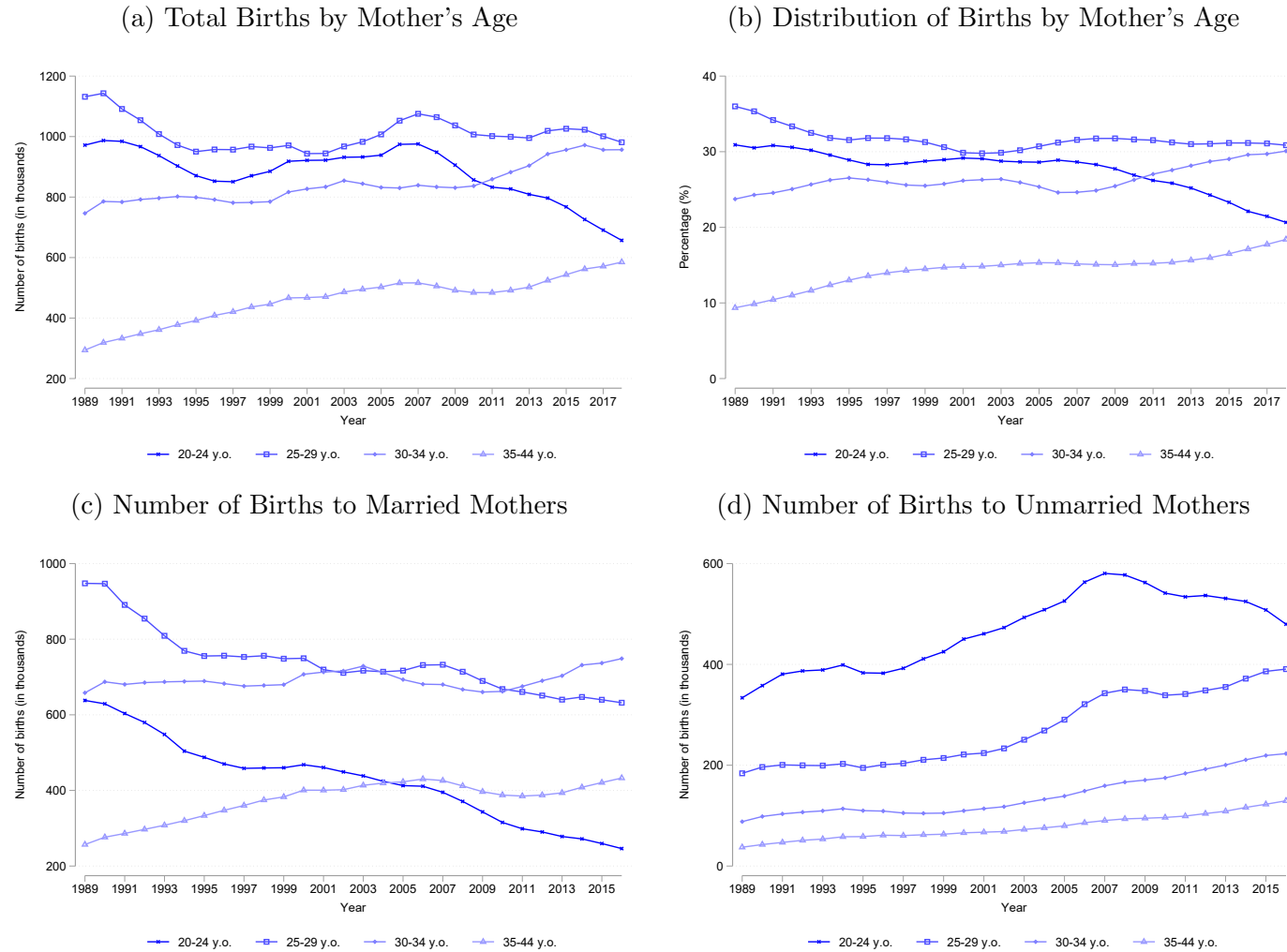
Figure 1: Geographical Variation



Notes: Panel (a) of this figure shows the geographic distribution of the cancer mortality rate (per 1,000) in 1996, our measure of exposure to the opioid epidemic. Panel (b) shows the geographic distribution of drug-induced mortality (per 10,000) between 1996 and 2022. Panel (c) shows the change in total fertility rates from women 15 to 44 between 1996 to 2018. Panel (d) shows the change in out migration probabilities from 1996 to 2018. The break points correspond to the quartiles of the variable's distribution. The sample is restricted to areas with more than 25,000 residents in 1996; these account for more than 99% of the total population. Commuting zones not included in the sample are white in the figure. We use geocoded National Vital Statistics System (NVSS) data to construct death counts; see Section III. for the ICD codes used. This figure is referenced in Section III.a

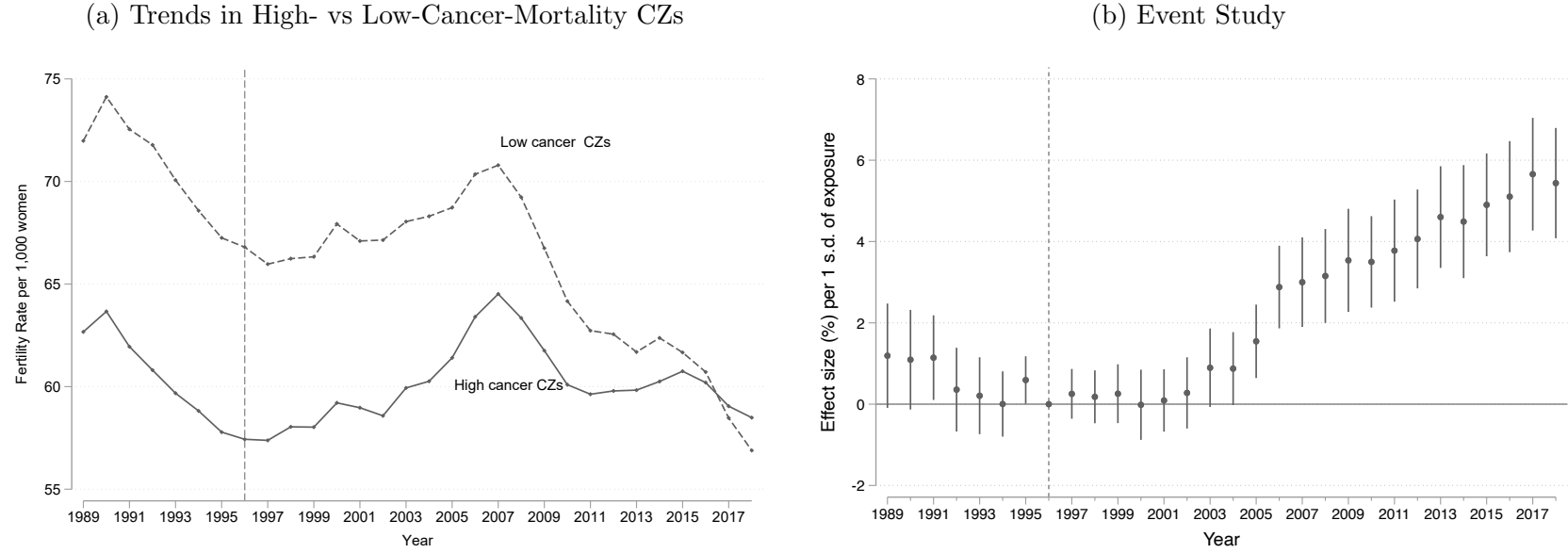


Figure 2: Evolution of Births by Age and Marital Status



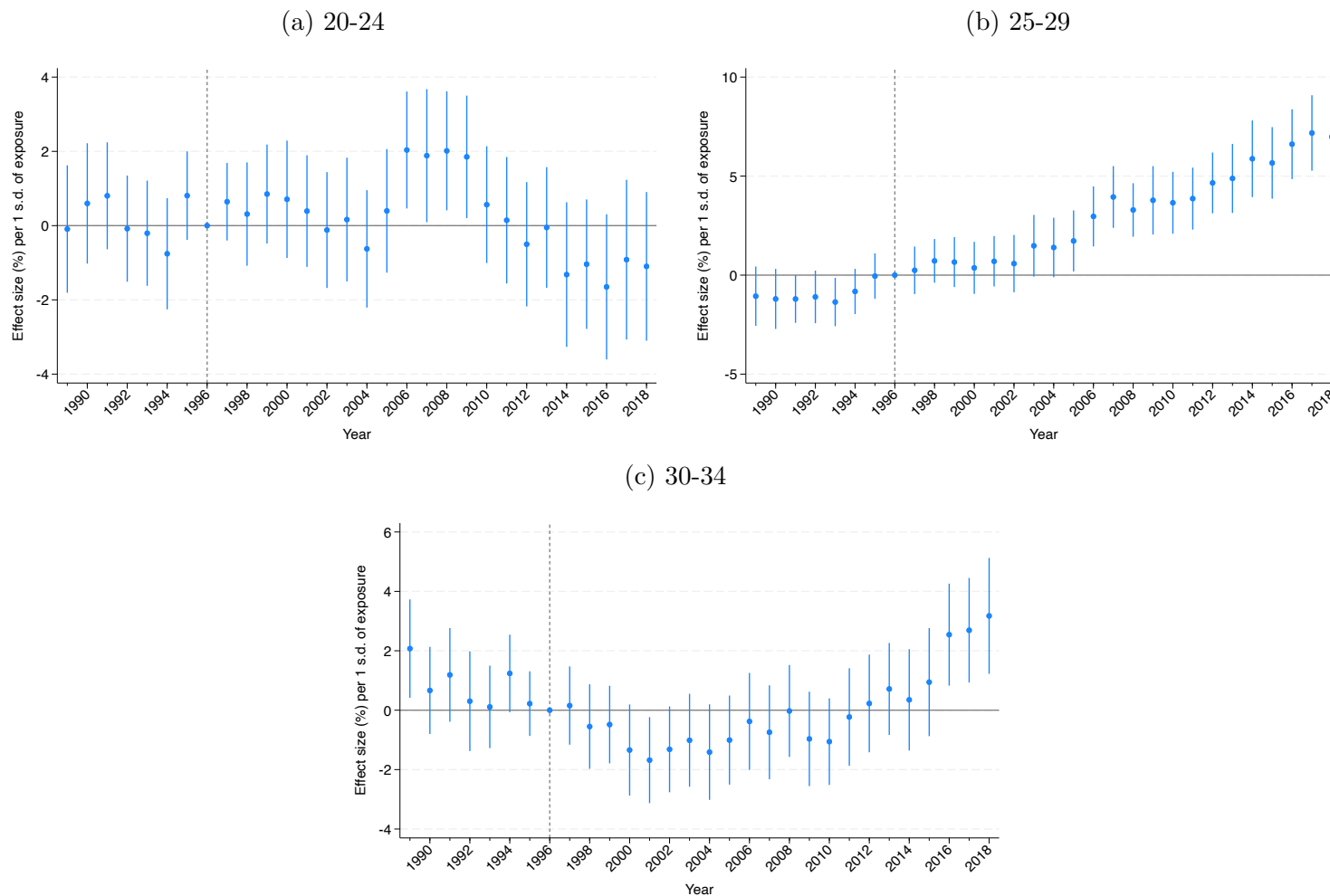
Notes: This figure presents trends in the number of births by age and marital status. Panel (a) shows the number of births to women aged 20 to 44. Panel (b) presents these data as shares, showing the proportion of births to mothers in each age subgroup among all mothers aged 20 to 44. Panels (c) and (d) display the evolution of the number of births by marital status for these age groups. These panels include data only through 2016, as that is the last year in which California birth certificates reported maternal marital status. This figure is referenced in Section III.b.

Figure 3: Effects of the Opioid Epidemic on Total Fertility



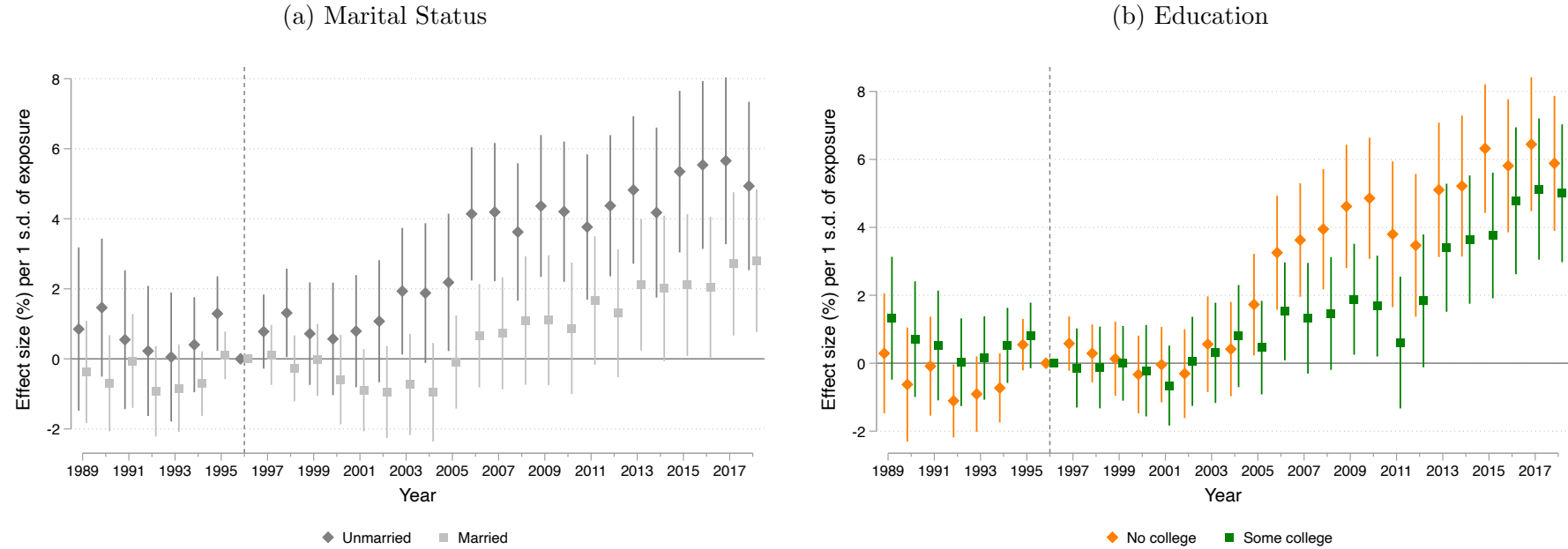
Notes: Total fertility is defined as the ratio of births to women aged 15 to 44 to the female population in this age group. Panel (a) shows the evolution of the fertility rate per 1,000 women in commuting zones (CZs) in the bottom (dashed lines) and top (solid lines) quartiles of cancer mortality prior to the launch of OxyContin. This comparison is based on a binary high-versus-low classification of CZs, with the outcome weighted by population. Panel (b) presents estimates of the effect of exposure to the opioid epidemic on fertility rates. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. This figure is referenced in Section V.a.

Figure 4: Exposure to the Opioid Epidemic and Fertility by Age Groups



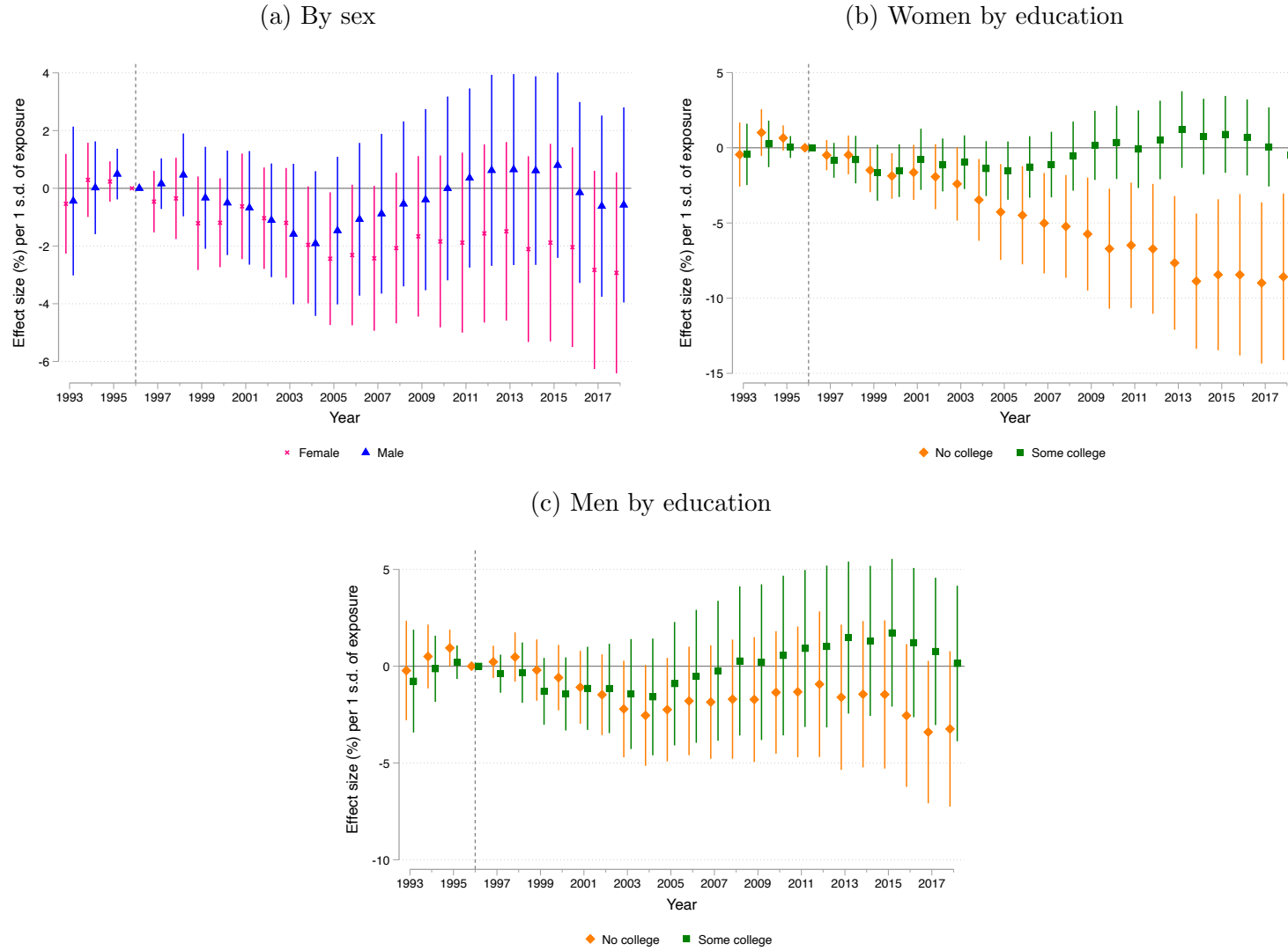
Notes: This figure presents the effects of exposure to the opioid epidemic on fertility rates by age group. The three age groups included account for 75% of births to women aged 15 to 44 in 1996. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section [V.a](#)

Figure 5: Effects of the Opioid Epidemic on Fertility by Marital Status and Education



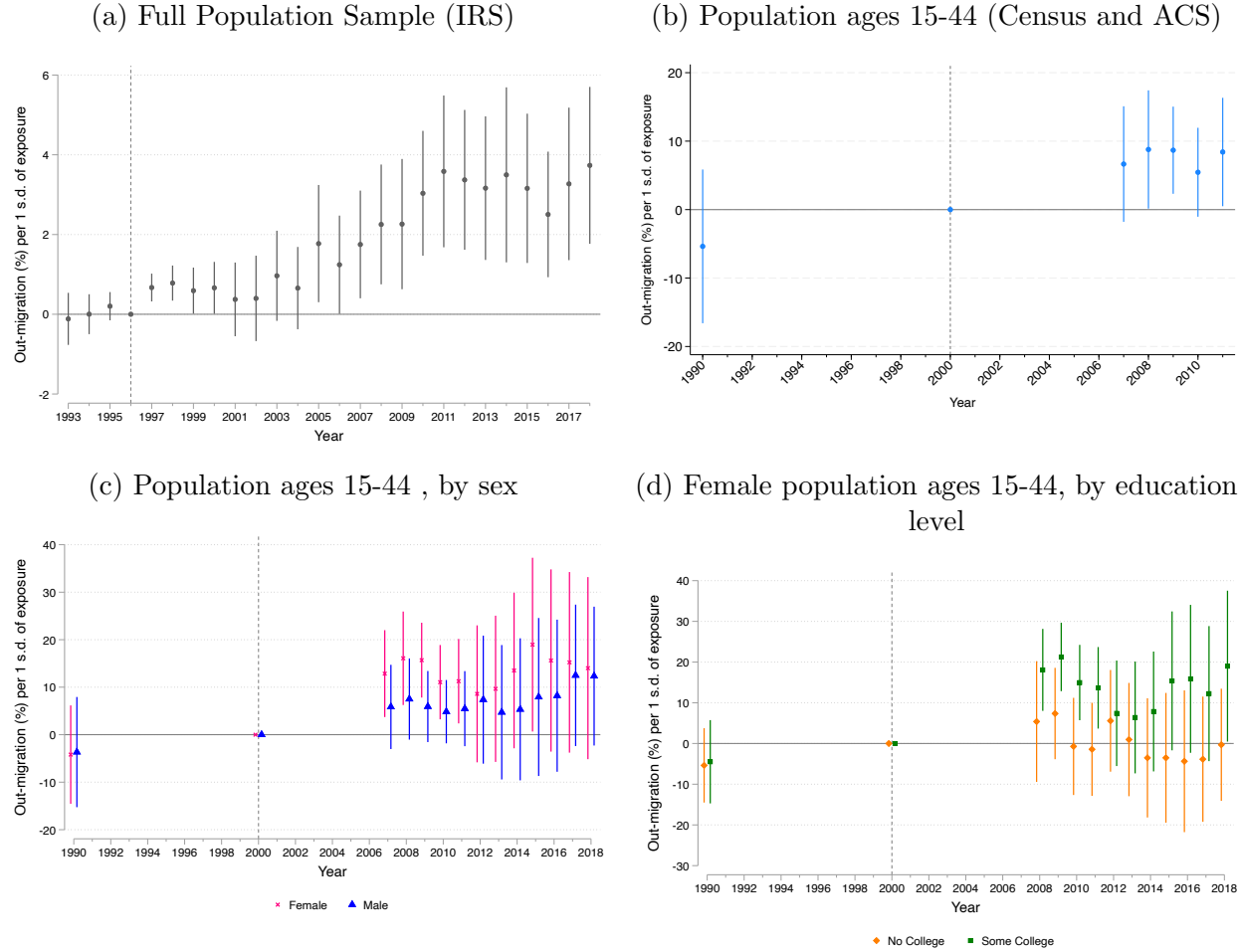
Notes: This figure shows the effects of exposure to the opioid epidemic on fertility rates by marital status (panel a) and educational level of the mother (panel b). The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. The married (unmarried) fertility rate is defined as the number of births to married (unmarried) individuals aged 15 to 44 years old divided by the population of married (unmarried) women in that age group. Similarly, fertility rates by education are the ratio of births to women with a no college (some college) to the number of women with no college (some college). Standard errors are clustered at the CZ level. This figure is referenced in Section V.a.

Figure 6: Exposure to the Opioid Epidemic and Employment Rates by Sex and Education



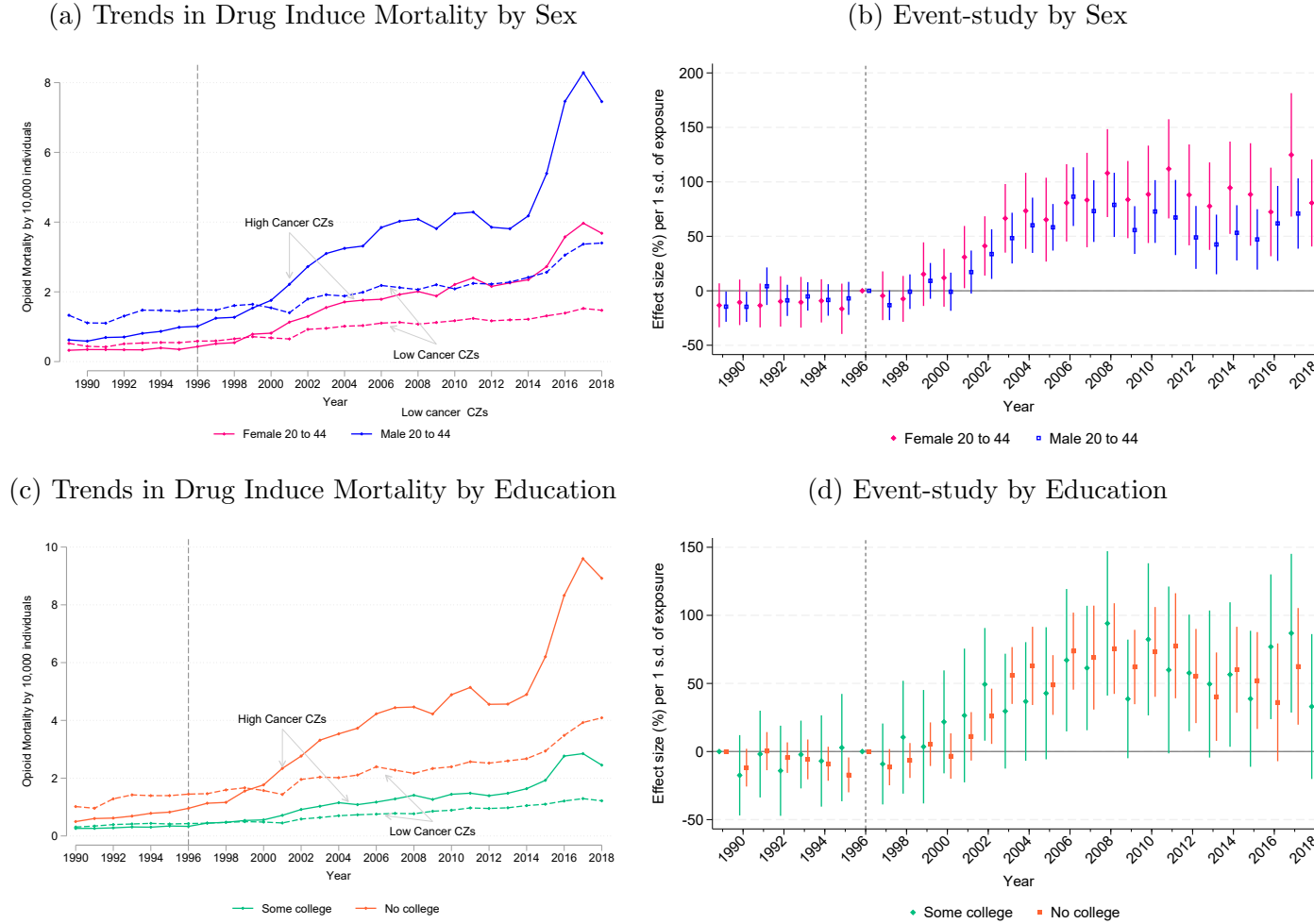
Notes: This figure presents the effects of exposure to the opioid epidemic on employment rates by sex and sex and education. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section VI.b

Figure 7: Effects of the Opioid Epidemic on Out-Migration



Notes: This figure presents the effects of exposure to the opioid epidemic on migration rates. Data from the IRS (panel a) covers inter-county internal migration for individuals of all ages and provides year to year moves. The IRS suppresses county-to-county migration flows below certain thresholds for confidentiality purposes; we therefore focus our analysis on county-pairs with 1996 migration rates in the top two quintiles. Data from the Census and ACS (panels b to d) are used to construct inter-CZ migration of individuals aged 15 to 44. Data coverage restricts the analysis to years 1990, 2000, and 2007 onward (future versions of this paper will present migration results to 2018). To improve precision in panel (d) we limit our sample to CZ pairs with above-median 2000 migration rates. The effect size (expressed as a percentage) corresponds to the scaled coefficients  $\phi_o$  from Equation (2). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates. Standard errors are clustered two-way at the origin-destination level. This figure is referenced in Section VI.c

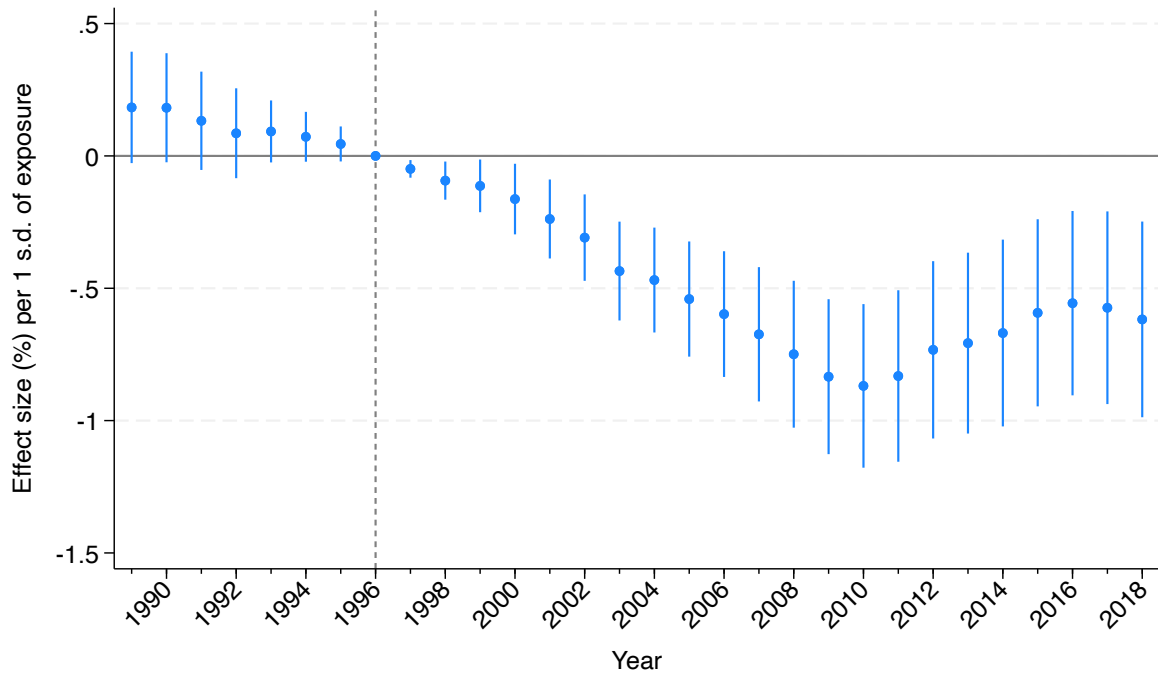
Figure 8: Exposure to the Opioid Epidemic and Drug Induced Mortality by Sex and Education



Notes: This figure shows the effects of the opioid epidemic on drug-induced mortality of women and men aged 20 to 44 years old. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section VI.d

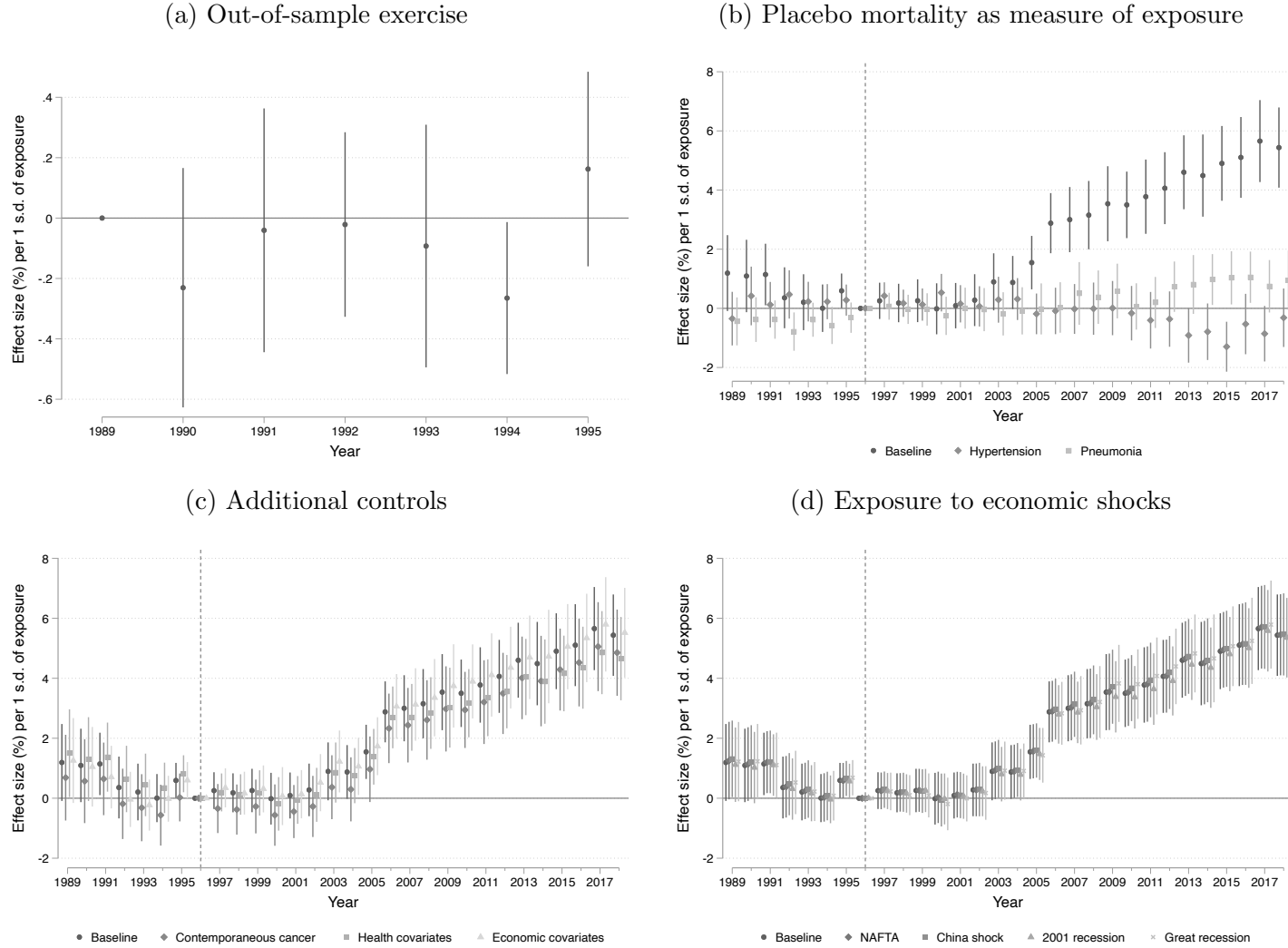


Figure 9: Share of women in the population of 20 to 44 years old



Notes: This figure shows the effects of the opioid epidemic on the share of the female population aged 20 to 44. Specifically, it presents the ratio of women in this age group to the total population aged 20 to 44. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. These estimates exploit continuous variation in cancer mortality in 1996 and control for baseline characteristics and state-by-year fixed effects. Standard errors are clustered at the CZ level. This figure is referenced in Section VI.d

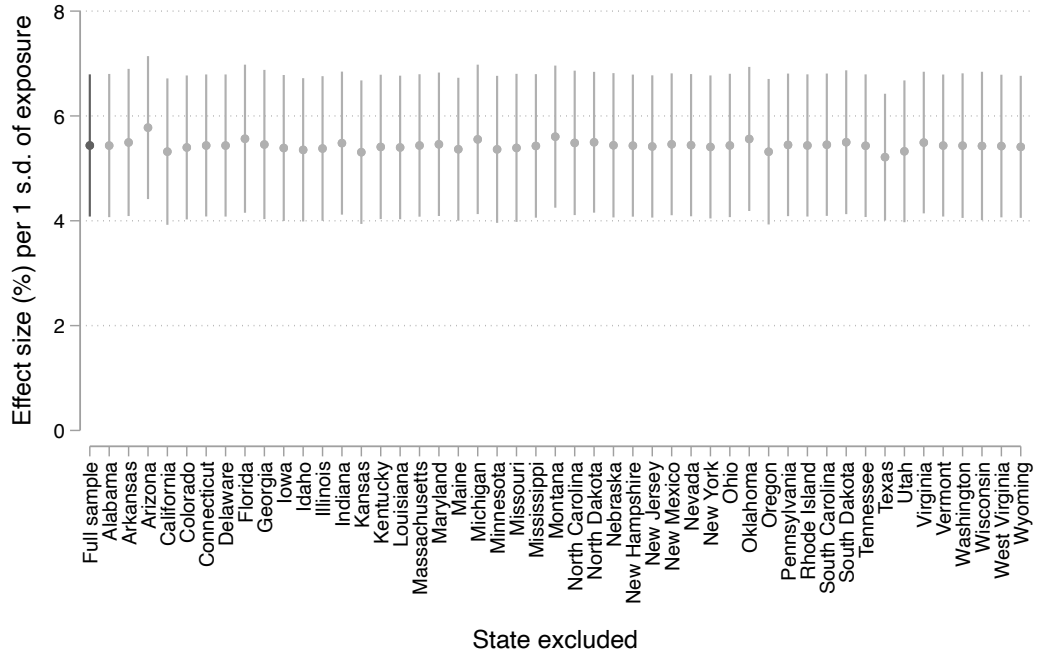
Figure 10: Robustness Checks - Total Fertility



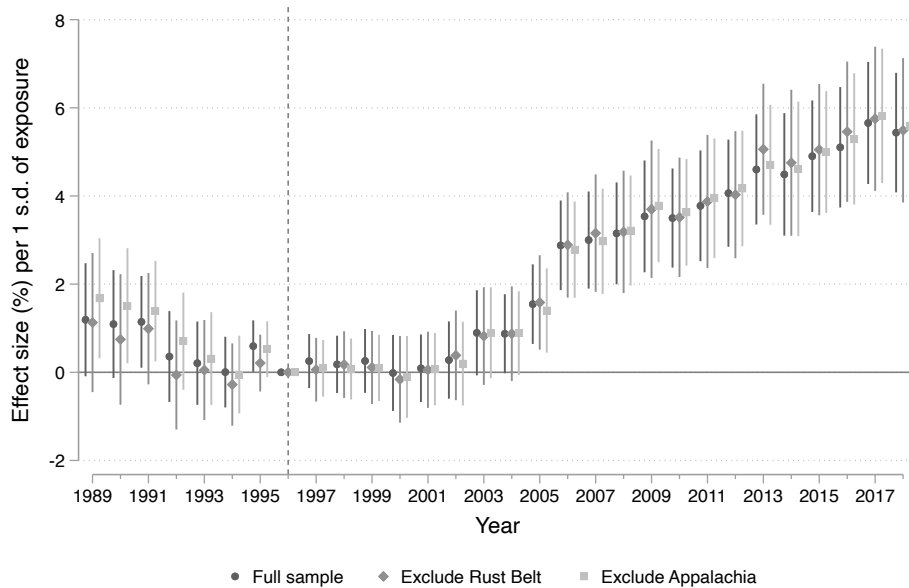
Notes: This figure presents robustness checks that probe the validity of our empirical strategy. The outcome of interest in each regression is the total fertility rate, i.e., the ratio of births to women aged 15 to 44 to the female population in this age group. Panel (a) shows an out-of-sample exercise where we use 1976 cancer mortality as our measure of exposure to the epidemic. Panel (b) presents placebo exercises where we use 1996 mortality from hypertension and pneumonia as the measure of exposure. Panel (c) expands the set of health and economic controls. For the regression in panels (c) and (d) each of these additional variables are interacted with year dummies. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. This figure is referenced in Section VII.

Figure 11: Estimation on Alternative Samples - Total Fertility

(a) Leave one state out - Year 2018



(b) Leave out specific regions



Notes: This figure presents estimations of the effects of exposure to the opioid epidemic on total fertility on alternative samples. Total fertility is defined as the ratio of births to women aged 15 to 44 to the female population in this age group. Panel (a) shows estimates leaving one state out at the time, coefficients correspond to 2018, the last year in the sample. Panel (b) presents estimates excluding all the commuting zones in the Appalachia and the Rust Belt regions. The effect size (expressed as a percentage) corresponds to the scaled coefficients from Equation (1). Specifically, we multiply the estimated coefficient by the standard deviation of 1996 cancer mortality rates and divide by the mean of the outcome variable in 1996. This figure is referenced in Section VII.

## X. Tables

Table 1: Summary Statistics

|                                                              | 1989-1996 |        |        | 2010-2018 |        |        |
|--------------------------------------------------------------|-----------|--------|--------|-----------|--------|--------|
|                                                              | Mean      | Median | SD     | Mean      | Median | SD     |
|                                                              | (1)       | (2)    | (3)    | (4)       | (5)    | (6)    |
| <b><i>Drug-induced mortality per 10,000 (ages 20-44)</i></b> |           |        |        |           |        |        |
| All                                                          | 0.0506    | 0.0416 | 0.0501 | 0.1950    | 0.1567 | 0.1738 |
| Male                                                         | 0.0680    | 0.0503 | 0.0788 | 0.2474    | 0.1961 | 0.2372 |
| Female                                                       | 0.0332    | 0.0197 | 0.0453 | 0.1412    | 0.1123 | 0.1434 |
| <b><i>Cancer mortality rate per 1,000</i></b>                |           |        |        |           |        |        |
|                                                              | 2.5191    | 2.5228 | 0.5386 | 2.4788    | 2.4954 | 0.5477 |
| <b><i>Fertility rate</i></b>                                 |           |        |        |           |        |        |
| <i>All ages (15-44)</i>                                      | 0.0636    | 0.0624 | 0.0102 | 0.0634    | 0.0627 | 0.0096 |
| 15-19                                                        | 0.0592    | 0.0593 | 0.0203 | 0.0309    | 0.0286 | 0.0143 |
| 20-24                                                        | 0.1282    | 0.1302 | 0.0361 | 0.1004    | 0.1003 | 0.0342 |
| 25-29                                                        | 0.1143    | 0.1115 | 0.0187 | 0.1215    | 0.1186 | 0.0222 |
| 30-34                                                        | 0.0649    | 0.0637 | 0.0146 | 0.0868    | 0.0860 | 0.0165 |
| 35-44                                                        | 0.0143    | 0.0136 | 0.0046 | 0.0215    | 0.0208 | 0.0063 |
| <b><i>Unmarried Women, 15-44</i></b>                         |           |        |        |           |        |        |
| 15-19                                                        | 0.0412    | 0.0390 | 0.0148 | 0.0475    | 0.0472 | 0.0131 |
| 20-24                                                        | 0.0401    | 0.0376 | 0.0163 | 0.0276    | 0.0257 | 0.0125 |
| 25-29                                                        | 0.0743    | 0.0714 | 0.0322 | 0.0783    | 0.0768 | 0.0309 |
| 30-34                                                        | 0.0529    | 0.0501 | 0.0195 | 0.0792    | 0.0769 | 0.0230 |
| 35-44                                                        | 0.0310    | 0.0291 | 0.0132 | 0.0538    | 0.0517 | 0.0157 |
|                                                              | 0.0080    | 0.0073 | 0.0044 | 0.0156    | 0.0148 | 0.0058 |
| <b><i>Married Women, 15-44</i></b>                           |           |        |        |           |        |        |
|                                                              | 0.0816    | 0.0791 | 0.0143 | 0.0851    | 0.0834 | 0.0129 |
| <b><i>Share of women</i></b>                                 |           |        |        |           |        |        |
| <i>Not married, 15-44</i>                                    | 0.4356    | 0.4311 | 0.0544 | 0.5759    | 0.5716 | 0.0581 |
| 15-19                                                        | 0.9340    | 0.9390 | 0.0302 | 0.9824    | 0.9860 | 0.0151 |
| 20-24                                                        | 0.6049    | 0.6015 | 0.1079 | 0.8104    | 0.8231 | 0.0832 |
| 25-29                                                        | 0.3364    | 0.3304 | 0.0675 | 0.5343    | 0.5356 | 0.0938 |
| 30-34                                                        | 0.2593    | 0.2547 | 0.0498 | 0.3859    | 0.3820 | 0.0767 |
| 35-44                                                        | 0.2369    | 0.2331 | 0.0419 | 0.3263    | 0.3234 | 0.0589 |
| <b><i>Population Share Female</i></b>                        |           |        |        |           |        |        |
| All ages (15-44)                                             | 0.4950    | 0.4977 | 0.0156 | 0.4847    | 0.4893 | 0.0213 |
| <b><i>Labor Force Participation</i></b>                      |           |        |        |           |        |        |
| Women (Ages 20-44)                                           | 0.7320    | 0.7382 | 0.0601 | 0.7483    | 0.7493 | 0.0586 |
| Men (Ages 20-44)                                             | 0.8675    | 0.8819 | 0.0607 | 0.8178    | 0.8360 | 0.0828 |

Notes: This table presents summary statistics for the main dependent variables and our measure of exposure to the opioid epidemic for the periods before and after the launch of OxyContin. This table is referenced in Section III.

Table 2: Marriage and Fertility Determinants in 2000

| Dependent variable         | Fertility rate: Unmarried<br>(1) | Fertility rate: Married<br>(2) | Fertility rate: No College<br>(3) | Fertility rate: Some College<br>(4) | Sh. Female 15-44 Married<br>(5) |
|----------------------------|----------------------------------|--------------------------------|-----------------------------------|-------------------------------------|---------------------------------|
| Sh. of population 50 - 64  | 0.132**<br>[0.0562]              | -0.176***<br>[0.0401]          | -0.140**<br>[0.0575]              | 0.01<br>[0.0379]                    | 0.652***<br>[0.246]             |
| Sh. of population over 66  | -0.08<br>[0.0562]                | 0.03<br>[0.0286]               | 0.03<br>[0.0472]                  | -0.001<br>[0.0329]                  | -0.23<br>[0.195]                |
| Sh. White                  | -0.0619***<br>[0.00933]          | 0.0105**<br>[0.00511]          | -0.0177**<br>[0.00808]            | -0.0238***<br>[0.00552]             | 0.131***<br>[0.0387]            |
| Sh. Hispanic               | 0.0667***<br>[0.0107]            | 0.0320***<br>[0.00837]         | 0.0548***<br>[0.0106]             | 0.0400***<br>[0.00660]              | 0.105***<br>[0.0397]            |
| Sh. Female                 | -0.294***<br>[0.0617]            | 0.127***<br>[0.0422]           | -0.09<br>[0.0651]                 | -0.02<br>[0.0409]                   | -0.703**<br>[0.348]             |
| Opioid mortality           | -0.01<br>[0.0326]                | -0.01<br>[0.0239]              | -0.0535*<br>[0.0319]              | 0.002<br>[0.0260]                   | -0.05<br>[0.137]                |
| Adult non-cancer mortality | 2.421***<br>[0.683]              | -1.859***<br>[0.441]           | -0.14<br>[0.678]                  | 0.33<br>[0.441]                     | 8.479***<br>[2.849]             |
| Emp rate: Males 15-44      | 0.129***<br>[0.0440]             | 0.143***<br>[0.0316]           | 0.145***<br>[0.0429]              | 0.0939***<br>[0.0319]               | 0.789***<br>[0.199]             |
| Emp rate: Females 15-44    | -0.11<br>[0.0661]                | -0.02<br>[0.0487]              | -0.07<br>[0.0585]                 | -0.05<br>[0.0384]                   | -0.754***<br>[0.221]            |
| Sh. Poverty                | 0.00<br>[0.0301]                 | 0.0475**<br>[0.0226]           | 0.04<br>[0.0294]                  | -0.0485**<br>[0.0214]               | -0.414***<br>[0.127]            |
| Ln Male Earnings           | 0.01<br>[0.00825]                | -0.01<br>[0.00639]             | -0.004<br>[0.00798]               | 0.004<br>[0.00612]                  | 0.02<br>[0.0332]                |
| Ln female Earnings         | -0.0155*<br>[0.00833]            | 0.01<br>[0.00656]              | 0.005<br>[0.00807]                | -0.005<br>[0.00619]                 | -0.036<br>[0.0330]              |
| Cancer mortality rate      | 0.0041<br>[0.00265]              | -0.0004<br>[0.00174]           | 0.0017<br>[0.00254]               | 0.0002<br>[0.00200]                 | 0.0005<br>[0.00939]             |
| Dep. var mean              | 0.045                            | 0.078                          | 0.088                             | 0.054                               | 0.537                           |
| N                          | 404                              | 404                            | 404                               | 404                                 | 404                             |

Notes: This table presents cross-sectional regressions of the fertility rate in 2000 at the CZ-level on various determinants. This table is referenced in Section III.

## A. Additional Figures and Tables

Table A1: Marriage and Fertility Determinants in 2010

| Dependent variable<br>Year: 2010 | Fertility rate: Unmarried<br>(1) | Fertility rate: Married<br>(2) | Fertility rate: No College<br>(3) | Fertility rate: Some College<br>(4) | Sh. Female 15-44 Married<br>(5) |
|----------------------------------|----------------------------------|--------------------------------|-----------------------------------|-------------------------------------|---------------------------------|
| Sh. of population 50 - 64        | 0.128***<br>[0.0451]             | -0.290***<br>[0.0402]          | -0.134*<br>[0.0716]               | 0.01<br>[0.0384]                    | 0.15<br>[0.205]                 |
| Sh. of population over 66        | -0.123***<br>[0.0401]            | 0.0998***<br>[0.0305]          | -0.0963*<br>[0.0499]              | 0.044<br>[0.0303]                   | -0.14<br>[0.140]                |
| Sh. White                        | -0.0378***<br>[0.00637]          | 0.00<br>[0.00433]              | -0.0341***<br>[0.00767]           | -0.0136***<br>[0.00500]             | 0.209***<br>[0.0237]            |
| Sh. Hispanic                     | 0.0743***<br>[0.00821]           | 0.00<br>[0.00506]              | 0.0598***<br>[0.00813]            | 0.0270***<br>[0.00654]              | 0.0761**<br>[0.0319]            |
| Sh. Female                       | -0.212***<br>[0.0544]            | 0.05<br>[0.0408]               | -0.136**<br>[0.0659]              | -0.04<br>[0.0471]                   | -0.457**<br>[0.214]             |
| Opioid mortality                 | -0.0139**<br>[0.00582]           | 0.0172***<br>[0.00499]         | 0.00<br>[0.00871]                 | -0.004<br>[0.00603]                 | 0.03<br>[0.0297]                |
| Adult non-cancer mortality       | 4.495***<br>[0.514]              | -0.704*<br>[0.402]             | 3.903***<br>[0.672]               | 1.641***<br>[0.413]                 | 9.895***<br>[1.937]             |
| Emp rate: Males 15-44            | -0.03<br>[0.0253]                | 0.0624***<br>[0.0194]          | -0.01<br>[0.0349]                 | 0.01<br>[0.0205]                    | -0.14<br>[0.103]                |
| Emp rate: Femles 15-44           | -0.0540**<br>[0.0241]            | -0.02<br>[0.0211]              | -0.03<br>[0.0334]                 | -0.0568**<br>[0.0229]               | -0.18<br>[0.110]                |
| Sh. Poverty                      | -0.0506***<br>[0.0188]           | 0.00<br>[0.0111]               | -0.0421*<br>[0.0226]              | -0.0742***<br>[0.0151]              | -0.705***<br>[0.0732]           |
| Ln Male Earnings                 | 0.00<br>[0.00600]                | 0.00<br>[0.00474]              | -0.009<br>[0.00748]               | 0.007<br>[0.00534]                  | -0.01<br>[0.0253]               |
| Ln female Earnings               | -0.003<br>[0.00602]              | 0.00<br>[0.00483]              | 0.009<br>[0.00759]                | -0.008<br>[0.00529]                 | 0.009<br>[0.0255]               |
| Cancer mortality rate            | 0.0013<br>[0.00204]              | -0.0029<br>[0.00194]           | 0.0020<br>[0.00240]               | -0.00401**<br>[0.00166]             | -0.0111<br>[0.00723]            |
| Dep. var mean                    | 0.050                            | 0.083                          | 0.092                             | 0.059                               | 0.443                           |
| N                                | 576                              | 576                            | 576                               | 576                                 | 576                             |